4.3 Formation of Niacin

Niacin (pyridine-3-carboxylic acid) is formed from the amino acid tryptophan. Niacin is the precursor for the synthesis of nicotinamide coenzymes. Only 3% of the tryptophan is converted to niacin in the liver. Maize eating population demonstrate niacin deficiency.

The steps involved in the synthesis of niacin (Figure 4.12) from tryptophan are:

1. Tryptophan is oxidized to N-formyl kynurenine.
2. N-formyl kynurenine transfers its formyl group to Tetrahydrofolate and results in the formation of Kynurenine
3. Kynurenine is hydroxylated at the 3rd position to form 3-hydroxy kynurenine.
4. Alanine is removed from 3-hydroxy kynurenine in the presence of pyridoxal phosphate to form 3-hydroxy anthranilic acid.
5. 3-hydroxy anthranilic acid is converted to quinolinic acid
6. Quinolinic acid on decarboxylation yields niacin.

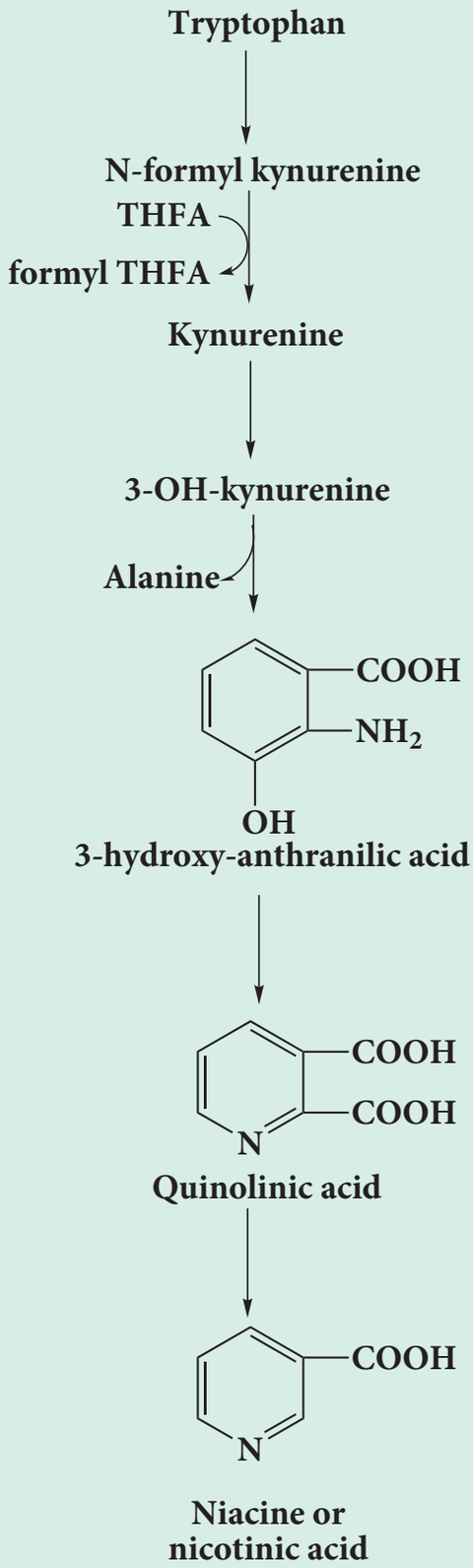

Figure 4.12. Biosynthesis of Niacin

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